

Beyond Linear Regression: The Journey to Support Vector Machines

Author: Soham Barua

Class: UG 4

Semester: 8

CRN: STAT001

Registration Number: 22214130001

Department: Department of Statistics, Presidency University

Instructor: Dr Shubhabrata Das, Department of Mathematics, Presidency University

Introduction

- Fisher's LDF has long been a popular method for classification tasks.
- Support Vector Machines (SVMs) were developed as a more powerful alternative and often provide superior classification performance.
- We first study the Maximal Margin Classifier, the basic idea behind SVMs.
- Since it requires perfectly separable data, its practical use is limited.
- This leads to the Support Vector Classifier, which allows some misclassification.
- Finally, we extend these ideas to the Support Vector Machine, capable of handling non-linear class boundaries.
- Together, these methods are commonly known as Support Vector Machines (SVMs).

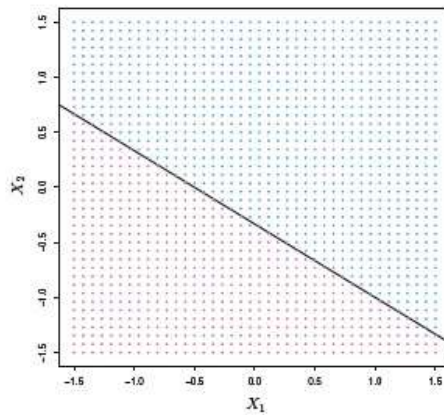
Defining the Hyperplane

- In a p -dimensional space, a hyperplane is a flat affine subspace, a subspace that does not need to pass through the origin, of dimension $p - 1$.
- It is defined by the equation,

$$\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p = 0$$

- Any $X = (X_1, X_2, \dots, X_p)$ satisfying this equation, are said to lie on the hyperplane.
- Points with positive and negative values of, $\beta_0 + \sum_{i=1}^p \beta_i X_i$, lie on opposite sides of the hyperplane.
- Thus, a hyperplane divides the feature space into two regions and serves as the foundation for classification methods such as SVMs.

FIGURE 1



The hyperplane $1 + 2X_1 + 3X_2 = 0$ is shown. The blue region is the set of points for which $1 + 2X_1 + 3X_2 > 0$ and the purple region is the set of points for which $1 + 2X_1 + 3X_2 < 0$.

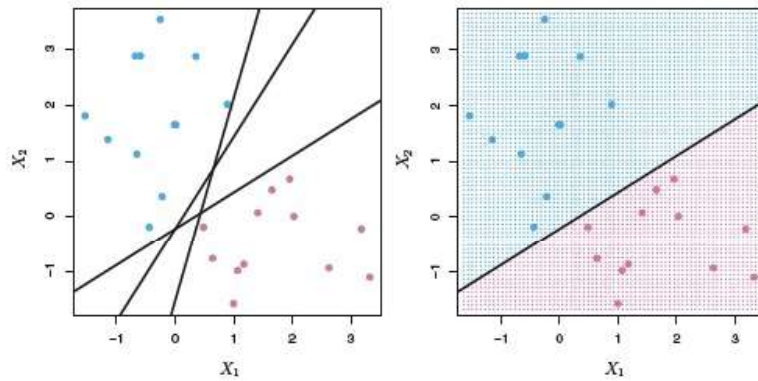
Classifying the Data using Hyperplanes

- Consider a training set containing observations x_i and class labels $y_i \in \{-1, +1\}$.
- The goal is to learn a decision function $f(x)$ that separates the two classes.
- Classification is performed using,

$$C(x) = \text{sign}(f(x))$$

- If $f(x) \geq 0$, the observation is assigned to class $+1$; otherwise, it is assigned to class -1 .
- A separating hyperplane uses this rule to divide the feature space into two region.
- While perfect separation is desirable, real-world data often contains overlapping classes, making some classification errors unavoidable.

FIGURE 2



Left: There are two classes of observations, shown in blue and purple, each of which has measurements on two variables. Three separating hyperplanes, out of many possible, are shown in black.

Right: A separating hyperplane is shown in black. The blue and purple grid indicates the decision rule made by a classifier based on this separating hyperplane.

Case 1: Data is linearly separable

- Suppose we have data such that we are able to construct a hyperplane that separates the training observations as per their respective class labels.
- Then, the separating hyperplane is characterized by,

$$\beta_0 + \sum_{j=1}^p \beta_j x_{ij} > 0 \text{ if } y_i = +1$$

and,

$$\beta_0 + \sum_{j=1}^p \beta_j x_{ij} < 0 \text{ if } y_i = -1$$

- Combining, we get,

$$(\beta_0 + \sum_{j=1}^p \beta_j x_{ij}) y_i > 0 \text{ for all } i = 1(1)n$$

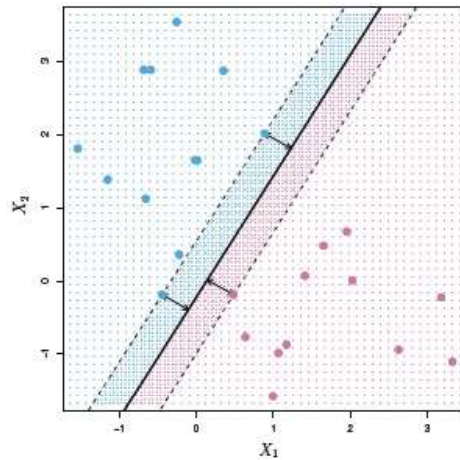
- Hence, if the data is linearly separable, then the separating hyperplane is the most natural classifier.
- But, infinite number of such hyperplanes exist. How do we get the best separating hyperplane?

The Maximal Margin Classifier

- Consider any such separating hyperplane.
- Let d_- be the shortest distance from the hyperplane to the nearest negative data point, and let d_+ be the shortest distance from the hyperplane to the nearest positive data point.
- Then, the width or the margin of the hyperplane is given by,

$$d = d_- + d_+$$
- Maximizing this distance(d), we shall get the optimal separating hyperplane. We call this the maximal margin classifier or hard margin classifier.

FIGURE 3



There are two classes of observations, shown in blue and in purple.

The maximal margin hyperplane is shown as a solid line. The margin is the distance from the solid line to either of the dashed lines. The two blue points and the purple point that lie on the dashed lines are the support vectors, and the distance from those points to the margin is indicated by arrows. The purple and blue grid indicates the decision rule made by a classifier based on this separating hyperplane.

What are Support Vectors?

- In case of linearly separable data, there exist β_0 and β such that,

$$\beta_0 + x_i^T \beta \geq +1 \text{ if } y_i = +1$$

$$\beta_0 + x_i^T \beta \leq -1 \text{ if } y_i = -1$$

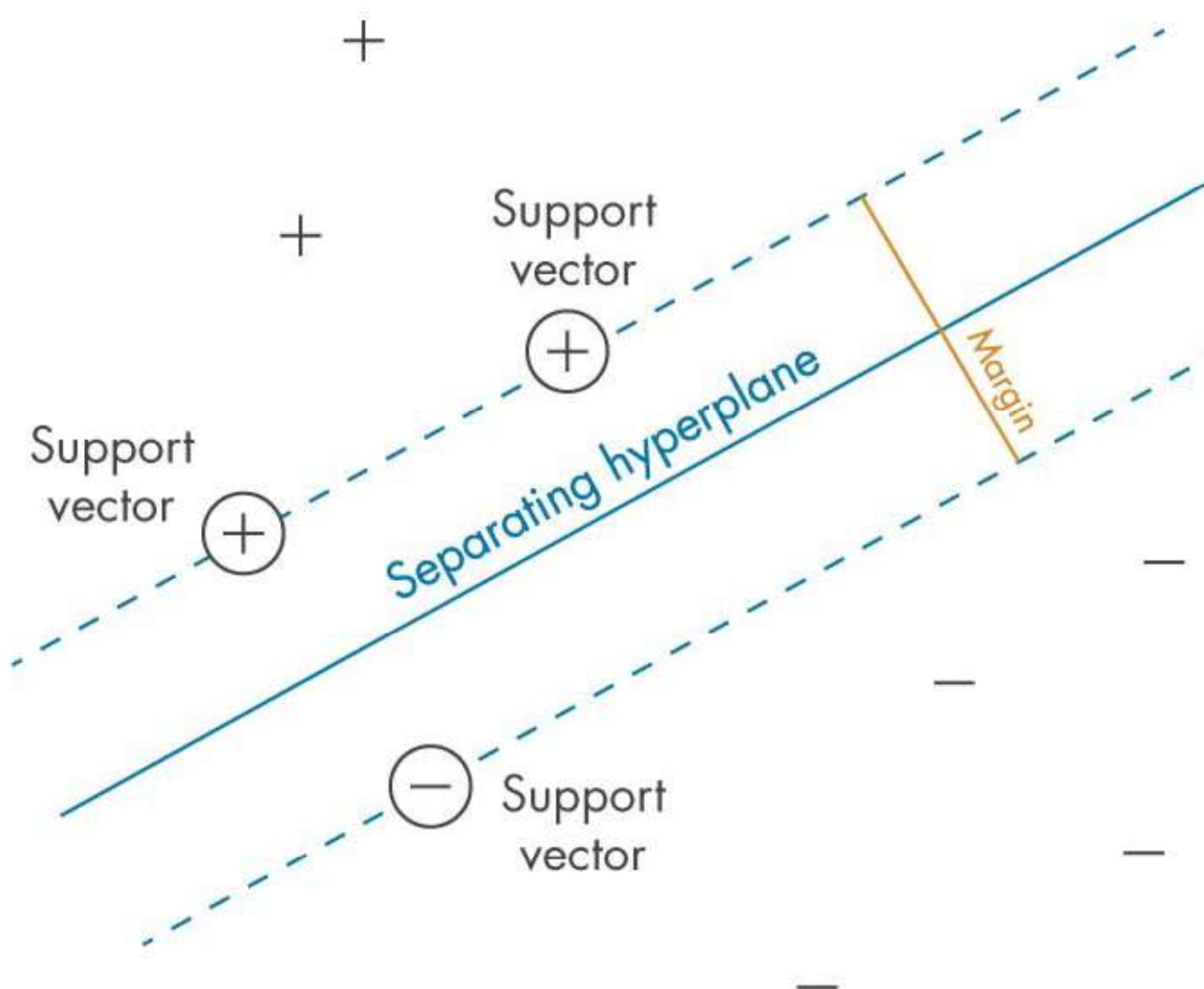
- Equality holds in either case when the data points lie exactly on the hyperplane, i.e., when,

$$\text{data points lie on } H_{+1} : (\beta_0 - 1) + x_i^T \beta = 0$$

or,

$$\text{data points lie on } H_{-1} : (\beta_0 + 1) + x_i^T \beta = 0$$

- Such data points are said to be support vectors.



Formulating the Margin

- Let x_{-1} be a point on H_{-1} , and x_{+1} be a point on H_{+1} .
- Then, we get,

$$\beta_0 + x_{-1}^T \beta = -1$$

and,

$$\beta_0 + x_{+1}^T \beta = +1$$

- Subtracting, we get,

$$x_{+1}^T \beta - x_{-1}^T \beta = 2$$

- Adding, we get,

$$\beta_0 = \frac{1}{2}(x_{+1}^T \beta + x_{-1}^T \beta)$$

- Consider another hyperplane, $\beta_0 + x^T \beta = 0$.
- Perpendicular distance of this hyperplane from $x_{-1} = d_- = \frac{|\beta_0 + x_{-1}^T \beta|}{\|\beta\|} = \frac{1}{\beta}$.
- Perpendicular distance of this hyperplane from $x_{+1} = d_+ = \frac{|\beta_0 + x_{+1}^T \beta|}{\|\beta\|} = \frac{1}{\beta}$.
- Hence, Margin of separating hyperplane $= d = d_- + d_+ = \frac{2}{\|\beta\|}$.
- Again, recall,

$$\beta_0 + x_i^T \beta \geq +1 \text{ if } y_i = +1$$

$$\beta_0 + x_i^T \beta \leq -1 \text{ if } y_i = -1$$

- Then, $(\beta_0 + x_i^T \beta)y_i \geq +1$ for any $i = 1(1)n$.
- $(\beta_0 + x_i^T \beta)y_i$, is called the margin of (x_i, y_i) with respect to hyperplane $\beta_0 + x^T \beta = 0$.
- Hence, x_i is a support vector if its margin is 1.

The Maximal Margin Classifier (Revisited) — Karush-Kuhn-Tucker Conditions

- The problem now is to maximize $\frac{2}{\|\beta\|}$, subject to, $y_i(\beta_0 + x_i^T \beta) \geq +1$.
- Or, minimize $\frac{1}{2}\|\beta\|^2$, subject to $y_i(\beta_0 + x_i^T \beta) \geq +1$.
- Using Lagrangian multipliers, we let,

$$\mathcal{L}(\beta_0, \beta, \alpha) = \frac{1}{2}\|\beta\|^2 - \sum_{i=1}^n \alpha_i \{y_i(\beta_0 + x_i^T \beta) - 1\}$$

where, $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$ is a vector of Lagrangian multipliers.

- We then minimize $\mathcal{L}$ with respect to β_0 and β (primal problem), and then maximize $\mathcal{L}^{min}$ with respect to α (dual problem).
- Solving the above two problems we get,

—

$$\hat{\beta} = \sum_{i \in sv} \hat{\alpha}_i y_i x_i$$

—

$$\hat{\beta}_0 = \frac{1}{|sv|} \sum_{i \in sv} \left(\frac{1 - y_i x_i^T \hat{\beta}}{y_i} \right)$$

- Optimal Hyperplane is,

—

$$\hat{\beta}_0 + \sum_{i \in sv} \hat{\alpha}_i y_i (x_i^T x_i)$$

- Maximal Margin is,

—

$$2 \left(\sum_{j \in sv} \hat{\alpha}_j \right)^{-\frac{1}{2}}$$

Why Hard Margin SVM fails?

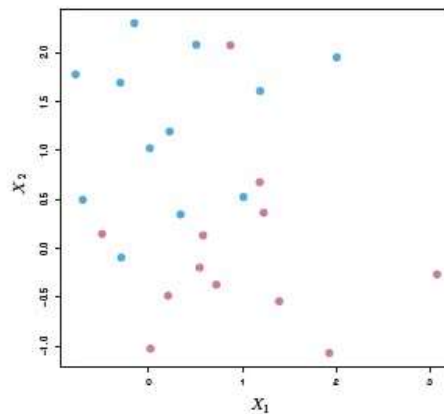
- The maximal margin classifier (hard margin SVM) works only when a separating hyperplane exists.
- Recall that hard margin SVM requires,

$$y_i(\beta_0 + x_i^T \beta) \geq 1; i = 1(1)n$$

which means every training observation must be classified correctly and must lie outside the margin.

- However, there are many situations where non-separability occurs, such as when classes are not linearly separable, when classes overlap heavily, or the data is noisy.
- Even when a separating hyperplane exists, using it may not be desirable.
- Hard margin SVM forces training error to be zero. As a consequence, the classifier becomes highly sensitive to individual observations and responds heavily to outliers.
- Even if all training observations are easily separable, but one observation lies close to the opposite class, then the hyperplane shifts to accomodate that single point, the margin shrinks substantially, and the decision boundary is rendered unstable.
- So, instead of demanding perfect classification, we may allow:
 - A few observations to be misclassified.
 - Some observations to lie inside the margin
- Main motivation behind soft margin SVM is,
 - Allow small violations $\implies$ More robust classifier with better generalization.
- Such a support vector classifier is called Soft Margin Classifier.

FIGURE 4



There are two classes of observations, shown in blue and in purple.

In this case, the two classes are not separable by a hyperplane, and so the maximal margin classifier cannot be used.

Slack Variables

- To allow violations of the margin constraints, introduce slack variables,

$$\xi_i \geq 0; i = 1(1)n$$

- The hard margin constraint,

$$y_i(\beta_0 + x_i^T \beta) \geq 1$$

is replaced by,

$$y_i(\beta_0 + x_i^T \beta) \geq 1 - \xi_i; i = 1(1)n$$

$$\iff y_i(\beta_0 + x_i^T \beta) + \xi_i \geq 1; i = 1(1)n$$

- **Interpretation of ξ_i :-**

- Case 1: $\xi_i = 0$

$$y_i(\beta_0 + x_i^T \beta) \geq 1$$

$$\iff \text{Observation is correctly classified}$$

- Case 2: $0 < \xi_i < 1$

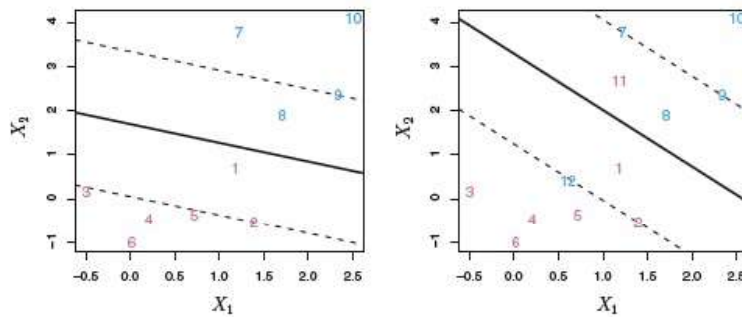
$$0 < y_i(\beta_0 + x_i^T \beta) < 1$$

$$\iff \text{Observation is correctly classified but lies inside the margin}$$

- Case 3: $\xi_i > 1$

$$y_i(\beta_0 + x_i^T \beta) < 0$$

$$\iff \text{Observation is misclassified}$$

FIGURE 5

Left: A support vector classifier was fit to a small data set. The hyperplane is shown as a solid line and the margins are shown as dashed lines.

Purple observations: Observations 3, 4, 5, and 6 are on the correct side of the margin, observation 2 is on the margin, and observation 1 is on the wrong side of the margin. *Blue observations:* Observations 7 and 10 are on the correct side of the margin, observation 9 is on the margin, and observation 8 is on the wrong side of the margin. No observations are on the wrong side of the hyperplane.

Right: Same as left panel with two additional points, 11 and 12. These two observations are on the wrong side of the hyperplane and the wrong side of the margin.

Soft Margin Optimization Problem

- The classifier must simultaneously,
 - Maximize the margin
 - Minimize the total margin violations.
- For the L_1 -norm soft margin, the objective function is,

$$\min_{\beta_0, \beta, \xi} \left[\frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n \xi_i \right]$$

subject to,

$$\xi_i \geq 0,$$

and,

$$y_i(\beta_0 + x_i^T \beta) \geq 1 - \xi_i; i = 1(1)n$$

- **Role of the regularization parameter C :-**

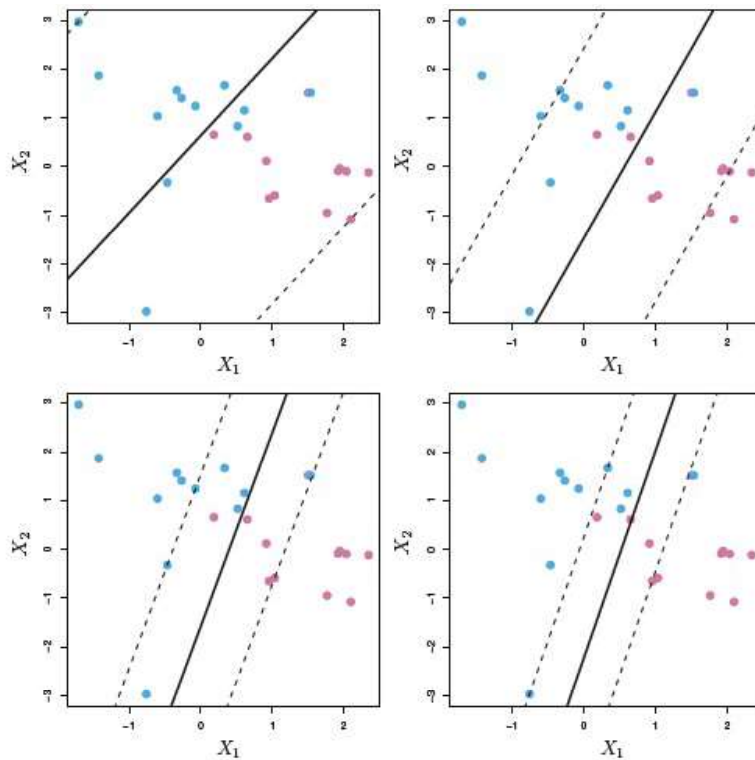
$$C > 0,$$

controls the trade off between $\frac{1}{2} \|\beta\|^2$ (large margin) and $\sum_{i=1}^n \xi_i$ (small classification errors).

- **Interpretation of C :-**

- Large $C \implies$ Misspecification is heavily penalized $\implies$ Smaller violations, narrower margin.
- Small $C \implies$ Misspecification is less penalized $\implies$ More violations, wider margin.

FIGURE 6



A support vector classifier was fit using four different values of the tuning parameter C . The largest value of C was used in the top left panel, and smaller values were used in the top right, bottom left, and bottom right panels. When C is large, then there is a high tolerance for observations being on the wrong side of the margin, and so the margin will be large. As C decreases, the tolerance for observations being on the wrong side of the margin decreases, and the margin narrows.

Finding the Optimal Solution

- After introducing Lagrange multipliers and solving the stationarity conditions,

$$\sum_{i=1}^n \alpha_i y_i = 0$$

$$\beta^{optimal} = \sum_{i=1}^n \alpha_i y_i x_i$$

and,

$$\alpha_i = C - \eta_i; i = 1(1)n$$

where,

$$\eta_i \geq 0 \forall i$$

- Substituting into the Lagrangian, we get,

$$\mathcal{L}^{min}(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (x_i^T x_j)$$

which is exactly the same dual objective function as in the hard margin case.

- **New constraint on α_i :-**

– Since,

$$\alpha_i = C - \eta_i$$

and,

$$\eta_i \geq 0,$$

we get,

$$0 \leq \alpha_i \leq C.$$

– This is the crucial difference from hard margin SVM.

Final Solution for Soft Margin SVM — Karush-Kuhn-Tucker Conditions

- The conditions here are,

- **Primal Feasibility,**

*

$$y_i(\beta_0 + x_i^T \beta) - (1 - \xi_i) \geq 0$$

*

$$\xi_i \geq 0$$

- **Dual Feasibility,**

*

$$\alpha_i \geq 0$$

*

$$\eta_i \geq 0$$

where, α_i corresponds to,

$$y_i(\beta_0 + x_i^T \beta) \geq (1 - \xi_i)$$

and, η_i corresponds to,

$$\xi_i \geq 0.$$

- **Complementary Slackness,**

*

$$\alpha_i [y_i(\beta_0 + x_i^T \beta) - (1 - \xi_i)] = 0$$

*

$$\xi_i(\alpha_i - C) = 0$$

- **Interpretation of the KKT Conditions,**

- From,

$$\xi_i(\alpha_i - C) = 0,$$

either,

$$\xi_i = 0,$$

or,

$$\alpha_i = C.$$

Therefore,

$$\xi_i > 0 \implies \alpha_i = C.$$

- Hence, any observation that violates the margin (or is misclassified), must satisfy,

$$\alpha_i = C.$$

Dual Optimization Problem

- The Dual Objective Function remains,

$$\mathcal{L}^{min}(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (x_i^T x_j),$$

or equivalently,

$$\mathcal{L}^{min}(\alpha) = 1_n^T \alpha - \frac{1}{2} \alpha^T H \alpha$$

- Constraints being,

$$\begin{aligned} \alpha^T y &= 0 \\ 0 &\leq \alpha_i \leq C \end{aligned}$$

- In vector form,

$$0 \leq \alpha \leq C 1_n$$

- The extra upper bound C is the only change in the dual optimization problem.

- **Why the upper bound matters?**

- The condition $\alpha_i \leq C$ limits the influence of any single observation.
- Therefore,
 - * outliers cannot dominate the solution.
 - * classifier becomes more robust.
 - * overfitting is reduced.

- **Limiting Case,**

- If

$$C \rightarrow \infty,$$

then,

$$0 \leq \alpha_i \leq C,$$

becomes,

$$\alpha_i \geq 0.$$

Hence,

$$C = \infty \implies \text{Hard Margin SVM.}$$

- Thus, hard margin SVM is a special case of soft margin SVM.

Optimization Solution

- After solving the dual problem and obtaining $\hat{\alpha}$, the optimal weight vector is,

$$\hat{\beta} = \sum_{i \in sv} \hat{\alpha}_i y_i x_i$$

- As in the hard margin case, here too only support vectors contribute because non-support vectors have $\hat{\alpha}_i = 0$.
- In soft margin SVM,

- A support vector satisfies,

$$y_i(\beta_0 + x_i^T \beta) = 1 - \xi_i$$

$$\iff y_i(\beta_0 + x_i^T \beta) - (1 - \xi_i) = 0$$

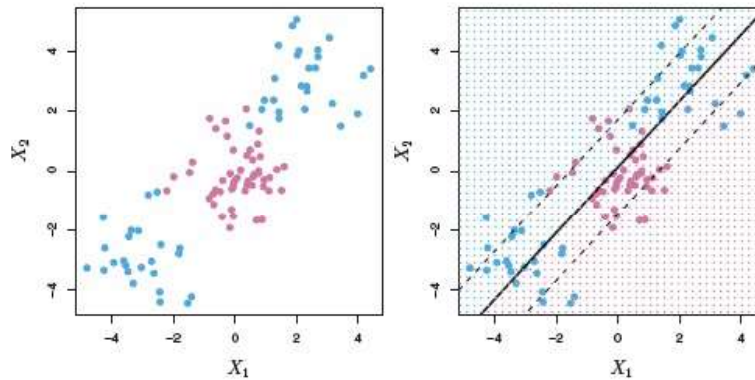
- These are exactly the observations for which the margin constraint is active.
- Thus,

$$sv = \{i : y_i(\beta_0 + x_i^T \beta) = 1 - \xi_i\}.$$

Case 2: Data is not linearly separable

- The maximal margin classifier is a very natural way to perform classification, if a separating hyperplane exists. However, there might be cases where there is no such separating hyperplane.
- In this case, we cannot separate the data exactly into two classes.
- However, we shall still try to develop a hyperplane that almost separates the classes.

FIGURE 7



Left: The observations fall into two classes, with a non-linear boundary between them.

Right: The support vector classifier seeks a linear boundary, and consequently performs very poorly.

Classification using Non-linear Decision Boundaries

- The support vector classifier (SVC) works well when the boundary separating two classes is approximately linear. However, many real-world classification problems involve non-linear decision boundaries, in which case a linear classifier may perform poorly.
- This situation is similar to what occurs in regression. When the relationship between predictors and the response is non-linear, we often improve performance by expanding the feature space using transformations of the predictors, such as quadratic or cubic terms.
- The same idea can be applied to support vector classifiers. Instead of fitting an SVC using only the original predictors,

$$X_1, X_2, \dots, X_p$$

we can enlarge the feature space by including transformed variables, for example,

$$X_1, X_1^2, X_2, X_2^2, \dots, X_p, X_p^2$$

- The support vector classifier is then fitted in this higher-dimensional space. The optimization problem becomes,

$$\max_{\beta_0, \beta_{11}, \beta_{12}, \dots, \beta_{p1}, \beta_{p2}, \epsilon_1, \dots, \epsilon_n} (Margin)$$

subject to,

1.

$$y_i \left(\beta_0 + \sum_{j=1}^p \beta_{j1} x_{ij} + \sum_{j=1}^p \beta_{j2} x_{ij}^2 \right) \geq (Margin)(1 - \epsilon_i)$$

2.

$$\sum_{i=1}^n \epsilon_i \leq C$$

3.

$$\epsilon \geq 0 \forall i$$

4.

$$\sum_{j=1}^p \sum_{k=1}^2 \beta_{jk}^2 = 1$$

- Although the separating boundary is linear in the enlarged feature space, it becomes non-linear in the original predictor space. For example, the decision boundary takes the form,

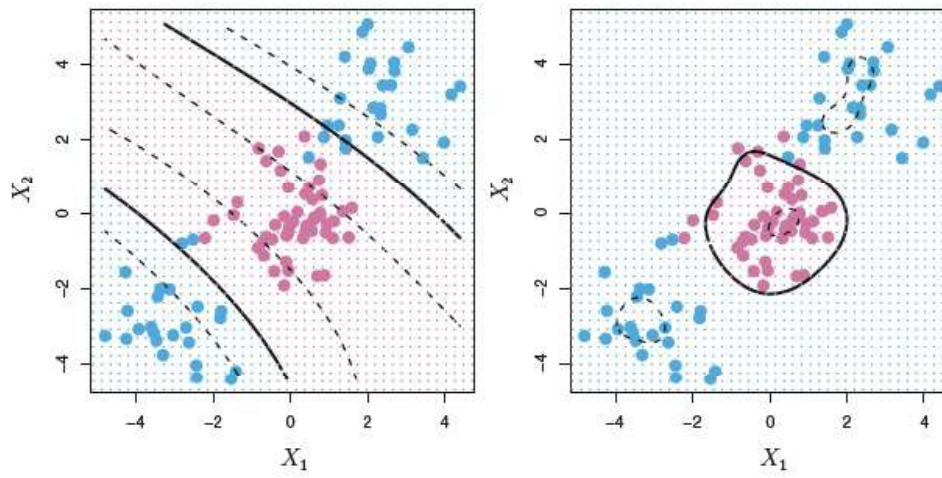
$$q(x) = 0$$

where, q is a quadratic polynomial. Consequently, the resulting classifier can capture curved boundaries that a standard linear SVC cannot.

Data is tangled in 2D → Map to a higher dimension → Find a linear split there

- These additions increase the flexibility of the classifier and allow it to model more complicated decision boundaries. However, a large number of transformed features can make computations expensive. The support vector machine (SVM) addresses this issue by enabling efficient computations in very high-dimensional feature spaces.

FIGURE 8



SVM using above methods. We can see that the data is better classified even when it is not linearly separable.

Conclusion: Why SVM is a powerful classification method?

- Support Vector Machines (SVMs) are particularly effective because they combine good predictive performance, flexibility, and strong theoretical foundations.
 - **Maximizes the Margin**
 - * SVM chooses the decision boundary with the largest possible margin between classes.
 - * A larger margin generally leads to better generalization on unseen data.
 - * This makes SVM less prone to overfitting than many other classifiers.
 - **Uses only important observations**
 - * The classifier is determined mainly by the support vectors (observations closest to the boundary).
 - * Points far from the boundary have little influence on the fitted model.
 - * This often makes SVM robust to noise in the bulk of the data.
 - **Handles non-linear boundaries**
 - * By transforming predictors into a higher-dimensional feature space, SVM can model complex non-linear decision boundaries.
 - * The resulting boundary may be highly flexible while still maintaining a large margin.
 - **Works well in high dimensions**
 - * SVM performs well when the number of predictors p is large, even when $p > n$.
 - * This is one reason SVM became popular in areas such as text classification, image recognition, and bioinformatics.
 - **Controls Model Complexity**
 - * The tuning parameter C controls the trade-off between:
 - a larger margin, and,
 - classification errors on the training data.
 - * This provides a principled way to balance bias and variance.
- SVM is attractive because it finds a maximum-margin classifier, can efficiently handle complex non-linear decision boundaries through kernels, uses only the most informative observations (support vectors), and often achieves excellent generalization performance even in high-dimensional settings.

References

- “An Introduction to Statistical Learning with Applications in R” by Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani
- “Support-Vector Networks” by Corinna Cortes, Vladimir Vapnik, AT&T Bell Labs., Holmdel, NJ 07733, USA
- <https://nl.mathworks.com/discovery/support-vector-machine.html>